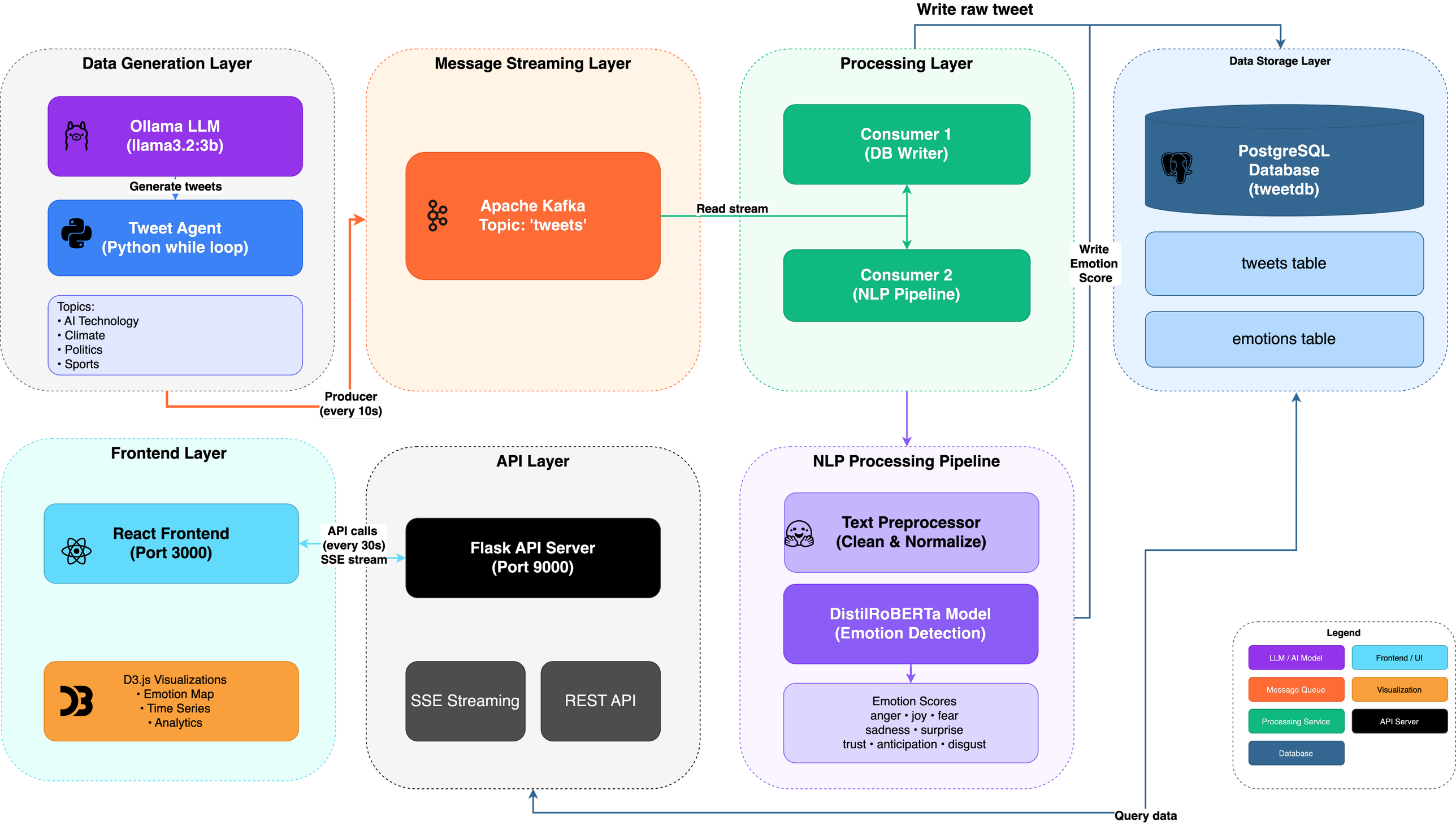
Write raw tweet
Data Generation Layer
Ollama LLM
(llama3.2:3b)
Generate tweets
Tweet Agent
(Python while loop)
Topics:
• AI Technology
• Climate
• Politics
• Sports
Producer
(every 10s)
Message Streaming Layer
Apache Kafka
Topic: 'tweets'
Read stream
Processing Layer
Consumer 1
(DB Writer)
Consumer 2
(NLP Pipeline)
Write
Emotion
Score
Data Storage Layer
PostgreSQL
Database
(tweetdb)
tweets table
emotions table
Frontend Layer
React Frontend
(Port 3000)
API calls
(every 30s)
SSE stream
D3.js Visualizations
• Emotion Map
• Time Series
• Analytics
API Layer
Flask API Server
(Port 9000)
SSE Streaming
REST API
NLP Processing Pipeline
Text Preprocessor
(Clean & Normalize)
DistilRoBERTa Model
(Emotion Detection)
Emotion Scores
anger • joy • fear
sadness • surprise
trust • anticipation • disgust
Legend
LLM / AI Model
Frontend / UI
Message Queue
Visualization
Processing Service
API Server
Database
Query data